

1. Question

Write a Java program to demonstrate exception handling using `ArrayIndexOutOfBoundsException` when accessing an invalid array index.

2. Objective

To write and execute a Java program that demonstrates **exception handling** using a try-catch block and handles the `ArrayIndexOutOfBoundsException`.

3. Steps for Execution

1. Create a Java file named `ArrayException.java`.
 2. Declare the package as `javapractice`.
 3. Create the class `ArrayException`.
 4. Declare an integer array containing three elements.
 5. Use a try block to access an array element.
 6. Access an invalid index, `arr[5]`.
 7. Since the array contains only three elements, an `ArrayIndexOutOfBoundsException` occurs.
 8. The catch block catches the exception.
 9. Display the appropriate error message.
 10. Compile and execute the program.
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4. Algorithm

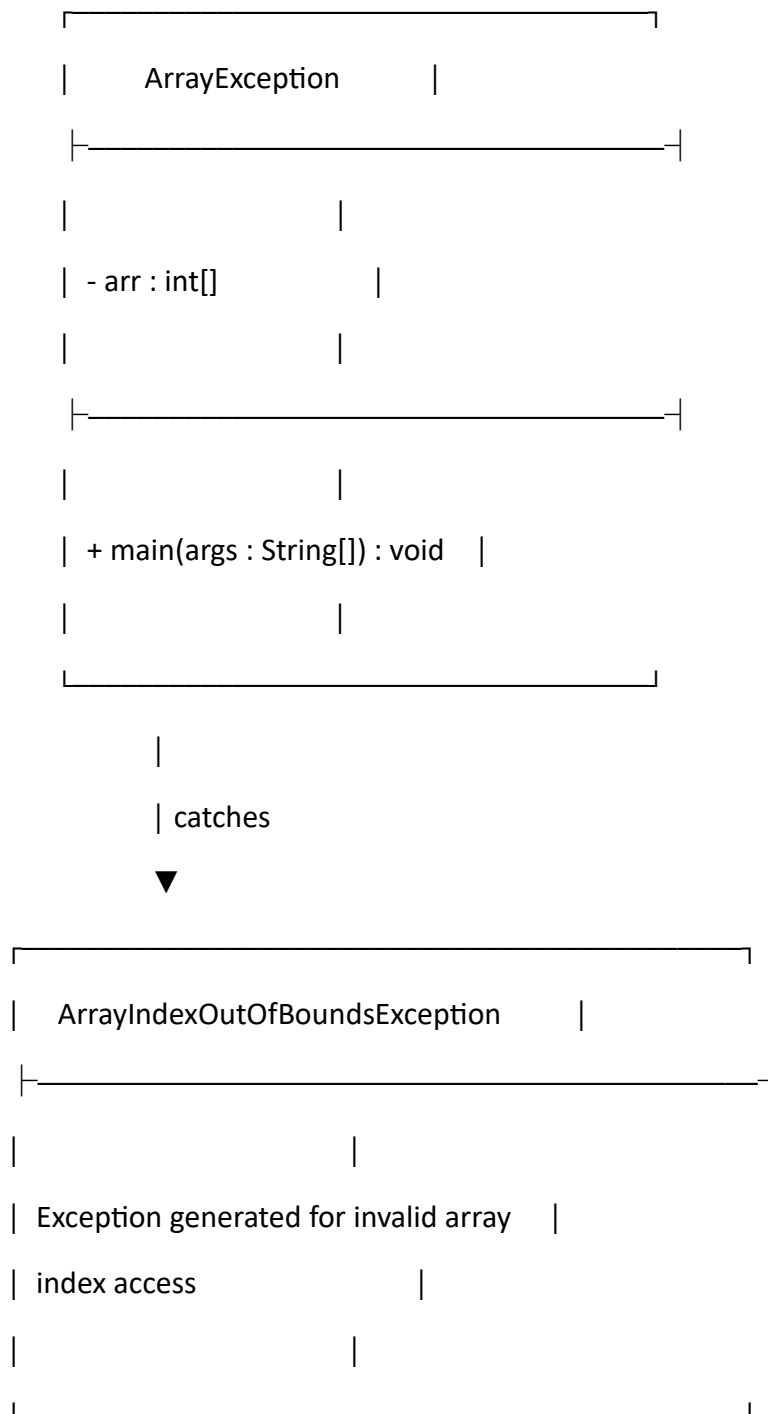
1. **Start**
2. Declare and initialize an integer array `arr = {10, 20, 30}`.
3. Start the try block.
4. Display "Accessing array element".
5. Try to access `arr[5]`.
6. Since index 5 is outside the valid range, an `ArrayIndexOutOfBoundsException` occurs.
7. Catch the exception using the catch block.

8. Display "Array index is out of bounds".

9. **Stop**

5. UML Class Diagram

Here is the **updated UML diagram** specifically matching your program:



6. Program Code

```
package javapractice;

public class ArrayException {
    public static void main(String[] args) {
        int arr[] = {10, 20, 30};

        try {
            System.out.println("Accessing array element");
            System.out.println(arr[5]);
        }
        catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Array index is out of bounds");
        }
    }
}
```

Output:

Accessing array element

Array index is out of bounds